REIN benchmark — detection results and denial-constraint attribution

beers, nasa, breast_cancer, bike, smartfactory, soilmoisture, adult · generated 2026-09-15 17:02

Beers section added on 2026-09-27; its Raha row is the first of the ten runs of 22 Sep (table below it). Tier labels renamed on 2026-09-27 (generic, oracle, domain → dirty profiled, clean profiled, hand written).

Does the beers constraint story transfer?

dataset	#DCs	mean DCs per column	table flagged	HoloClean F1	largest single ΔF1	cell_baseline	F1 – baseline
nasa	4	2.80 (max 4)	69.5%	0.1589	+0.0403	0.1500	+0.0090
breast_cancer	8	3.33 (max 6)	67.8%	0.1316	+0.0100	0.1399	-0.0083
bike	14	3.60 (max 11)	67.3%	0.2155	+0.0403	0.2079	+0.0076
smartfactory	18	6.63 (max 16)	56.5%	0.0536	+0.0020	0.0311	+0.0225
soilmoisture	0	—	—	—	—	0.4607	—
adult	14	5.07 (max 11)	98.1%	0.0050	+0.0003	0.0050	+0.0000

mean DCs per column is what decides whether a leave-one-out ablation can say anything. Near 1, a constraint owns its columns outright, so removing it removes them and the gap it leaves is interpretable — that is the beers case. Well above 1, the same columns are named by several constraints, removing any one changes almost nothing, and the ablation is blind by construction.

beers

metric	value
rows (N)	2,410
dimension (d)	11
cells	26,510
injected errors	4,362
columns with errors (d_err)	5 of 11
dirty row fraction	1.0000
cell error rate (eps)	0.1645
cell_baseline = 2eps/(1+eps)	0.2826
Rousseeuw prediction	0.8616
denial constraints	3

Detection results

detector	precision	recall	F1	F1 – baseline	#detections	runtime (s)
GreatExpectations (dirty profiled)	0.9441	0.2595	0.4071	0.1245	1,199	0.02
GreatExpectations (clean profiled)	0.9844	0.9709	0.9776	0.6950	4,302	0.03
GreatExpectations (hand written)	0.9849	1.0000	0.9924	0.7098	4,429	0.02
HoloClean	0.4393	0.5862	0.5022	0.2196	5,821	0.70
Raha	1.0000	0.9106	0.9532	0.6706	3,972	5.15

F1 – baseline compares each detector against the trivial detector that flags every cell (cell_baseline = 0.2826). Green beats it by more than 0.02, red is at or below it.

Raha — the ten runs of 22 Sep 2026

run	time	precision	recall	F1	#detections	TP	FP	FN	runtime (s)
1 (reported)	11:56:15	1.0000	0.9106	0.9532	3,972	3972	0	390	5.15
2	11:56:20	0.9984	1.0000	0.9992	4,369	4362	7	0	4.91
3	11:56:24	0.9984	1.0000	0.9992	4,369	4362	7	0	4.86
4	11:56:29	0.9785	1.0000	0.9891	4,458	4362	96	0	4.91
5	11:56:34	0.9984	1.0000	0.9992	4,369	4362	7	0	4.91

6	11:56:40	0.9984	1.0000	0.9992	4,369	4362	7	0	5.22
7	11:56:45	0.9979	1.0000	0.9990	4,371	4362	9	0	5.57
8	11:56:50	1.0000	0.9741	0.9869	4,249	4249	0	113	5.26
9	11:56:56	0.9984	1.0000	0.9992	4,369	4362	7	0	5.13
10	11:57:01	0.9979	1.0000	0.9990	4,371	4362	9	0	5.35
mean		0.9966	0.9885	**0.9923**	4,326.6				5.13
sd		0.0064	0.0285	**0.0145**	134.1				0.23

One process, ten runs, strategy results loaded from the existing cache (log `detection_logs/logs_raha_20260922-115609.txt`). The detection table above reports run 1. The first run differs from the other nine because it is the first draw of the seeded random stream. A run that rebuilds the strategy cache gives 3,922 detections (F1 0.9469) for that first run instead of 3,972 (F1 0.9532), because the strategies then come in a different order; runs 2–10 are unaffected.

HoloClean — per-constraint ablation

k	constraint (EQ → IQ)	cells	% rows	F1 alone	F1 without	ΔF1	unique TP
1	ounces → beer_name	4,810	99.8%	0.5244	0.0566	+0.4456	2,405
3	brewery_name, abv → city	342	4.7%	0.0234	0.5057	-0.0035	43
2	brewery_name, abv → ibu	771	10.7%	0.0425	0.5171	-0.0149	97

All 3 constraints together: P=0.4393, R=0.5862, **F1=0.5022**, flagging **22.0%** of the table. The trivial flag-everything detector scores 0.2826 (+0.2196).

HoloClean touches 6 of the 11 columns. Flagging *every* cell of just those columns — ignoring the constraints entirely and keeping only the column selection — scores F1=0.4500, against HoloClean's 0.5022 (+0.0522 for the row-level filtering the constraints actually perform).

nasa

metric	value
rows (N)	1,503
dimension (d)	6
cells	9,018
injected errors	731
columns with errors (d_err)	5 of 6
dirty row fraction	0.4864
cell error rate (eps)	0.0811
cell_baseline = 2eps/(1+eps)	0.1500
Rousseeuw prediction	0.3978
denial constraints	4

Detection results

detector	precision	recall	F1	F1 – baseline	#detections	runtime (s)
GreatExpectations (dirty profiled)	1.0000	0.0410	0.0788	-0.0711	30	0.01
GreatExpectations (clean profiled)	1.0000	0.5253	0.6888	0.5388	384	0.02
GreatExpectations (hand written)	1.0000	0.9302	0.9639	0.8139	680	0.01
HoloClean	0.0887	0.7606	0.1589	0.0090	6,265	0.32
Raha	0.2758	0.9713	0.4297	0.2797	2,574	1.79

F1 – baseline compares each detector against the trivial detector that flags every cell (cell_baseline = 0.1500). Green beats it by more than 0.02, red is at or below it.

HoloClean — per-constraint ablation

k	constraint (EQ → IQ)	cells	% rows	F1 alone	F1 without	ΔF1	unique TP
2	angle, chord_length → velocity	3,891	86.3%	0.1077	0.1186	+0.0403	156
1	angle → chord_length	2,844	94.6%	0.1365	0.1189	+0.0401	159
3	angle, chord_length, velocity → thickness	3,292	54.8%	0.0805	0.1228	+0.0361	136

4	angle, chord_length, velocity + 1 more → sound_pressure_level	4,670	62.1%	0.0089	0.1912	-0.0322	4

All 4 constraints together: P=0.0887, R=0.7606, **F1=0.1589**, flagging **69.5%** of the table. The trivial flag-everything detector scores 0.1500 (+0.0090).

HoloClean touches 5 of the 6 columns. Flagging *every* cell of just those columns — ignoring the constraints entirely and keeping only the column selection — scores F1=0.1438, against HoloClean's 0.1589 (+0.0151 for the row-level filtering the constraints actually perform).

breast_cancer

metric	value
rows (N)	699
dimension (d)	12
cells	8,388
injected errors	631
columns with errors (d_err)	12 of 12
dirty row fraction	0.6481
cell error rate (eps)	0.0752
cell_baseline = 2eps/(1+eps)	0.1399
Rousseeuw prediction	0.6088
denial constraints	8

Detection results

detector	precision	recall	F1	F1 – baseline	#detections	runtime (s)
GreatExpectations (dirty profiled)	0.9500	0.3914	0.5544	0.4145	260	0.02
GreatExpectations (clean profiled)	0.9553	0.4406	0.6030	0.4631	291	0.02
GreatExpectations (hand written)	0.9522	0.6941	0.8029	0.6630	460	0.01
HoloClean	0.0731	0.6593	0.1316	-0.0083	5,691	0.30
Raha	0.8025	0.5024	0.6179	0.4780	395	2.01

F1 – baseline compares each detector against the trivial detector that flags every cell (cell_baseline = 0.1399). Green beats it by more than 0.02, red is at or below it.

HoloClean — per-constraint ablation

k	constraint (EQ → IQ)	cells	% rows	F1 alone	F1 without	ΔF1	unique TP
8	class → Clump Thickness	1,398	100.0%	0.0867	0.1216	+0.0100	78
4	Mitoses, Bland Chromatin, Uniformity of Cell Size + 1 more → Single Epithelial Cell Size	1,940	55.5%	0.0685	0.1241	+0.0075	33
1	Mitoses → Bland Chromatin	1,394	99.7%	0.0998	0.1259	+0.0057	22
3	Mitoses, Bland Chromatin, Uniformity of Cell Size → Marginal Adhesion	2,056	73.5%	0.0722	0.1261	+0.0055	26
7	Uniformity of Cell Size, Normal Nucleoli, class → Uniformity of Cell Shape	2,504	89.6%	0.0906	0.1266	+0.0050	72
2	Mitoses, Bland Chromatin → Uniformity of Cell Size	1,938	92.4%	0.0903	0.1290	+0.0026	10
5	Mitoses, Bland Chromatin, Uniformity of Cell Size + 2 more → Normal Nucleoli	1,626	38.8%	0.0523	0.1304	+0.0012	5
6	Bland Chromatin, Uniformity of Cell Size, Normal Nucleoli → class	1,352	48.4%	0.0262	0.1318	-0.0002	0

All 8 constraints together: P=0.0731, R=0.6593, **F1=0.1316**, flagging **67.8%** of the table. The trivial flag-everything detector scores 0.1399 (-0.0083).

HoloClean touches 9 of the 12 columns. Flagging *every* cell of just those columns — ignoring the constraints entirely and keeping only the column selection — scores F1=0.1370, against HoloClean's 0.1316 (-0.0054 for the row-level filtering the constraints actually perform).

bike

metric	value
rows (N)	17,379
dimension (d)	17

cells	295,443
injected errors	34,274
columns with errors (d_err)	16 of 17
dirty row fraction	0.9437
cell error rate (eps)	0.1160
cell_baseline = 2eps/(1+eps)	0.2079
Rousseeuw prediction	0.8771
denial constraints	14

Detection results

detector	precision	recall	F1	F1 – baseline	#detections	runtime (s)
GreatExpectations (dirty profiled)	0.0000	0.0000	0.0000	-0.2079	0	0.23
GreatExpectations (clean profiled)	0.0000	0.0000	0.0000	-0.2079	0	0.25
GreatExpectations (hand written)	1.0000	0.7074	0.8287	0.6208	24,247	0.14
HoloClean	0.1263	0.7333	0.2155	0.0076	198,979	8.30
Raha	0.6709	0.9004	0.7689	0.5610	45,999	160.11

F1 – baseline compares each detector against the trivial detector that flags every cell (cell_baseline = 0.2079). Green beats it by more than 0.02, red is at or below it.

HoloClean — per-constraint ablation

k	constraint (EQ → IQ)	cells	% rows	F1 alone	F1 without	ΔF1	unique TP
12	workingday → holiday	32,976	94.9%	0.2250	0.1752	+0.0403	7,555
1	hum → season	33,086	95.2%	0.2117	0.2062	+0.0093	1,574
2	hum, season → hr	39,870	76.5%	0.1224	0.2109	+0.0046	648
3	hum, season, hr → weekday	31,372	45.1%	0.0230	0.2128	+0.0027	369
9	season, weekday, atemp → mnth	32,840	47.2%	0.1729	0.2140	+0.0015	505
11	season, hr, weekday + 1 more → workingday	2,825	3.3%	0.0102	0.2142	+0.0013	166
6	hum, season, hr + 1 more → casual	13,925	16.0%	0.0078	0.2146	+0.0009	137
10	season, weekday, mnth → instant	52,324	75.3%	0.2480	0.2150	+0.0005	2,403
7	hum, season, weekday + 1 more → temp	8,000	9.2%	0.0063	0.2151	+0.0005	95
4	hum, season, hr + 1 more → weathersit	8,300	9.6%	0.0090	0.2153	+0.0002	167
5	hum, season, hr + 2 more → windspeed	9,210	8.8%	0.0062	0.2171	-0.0016	110
8	season, weekday, temp → atemp	34,216	49.2%	0.0411	0.2201	-0.0046	550
14	cnt → registered	34,554	99.4%	0.0254	0.2272	-0.0117	874
13	hr, casual → cnt	44,802	85.9%	0.0000	0.2302	-0.0147	0

All 14 constraints together: P=0.1263, R=0.7333, **F1=0.2155**, flagging **67.3%** of the table. The trivial flag-everything detector scores 0.2079 (+0.0076).

HoloClean touches 15 of the 17 columns. Flagging *every* cell of just those columns — ignoring the constraints entirely and keeping only the column selection — scores F1=0.2205, against HoloClean's 0.2155 (-0.0050 for the row-level filtering the constraints actually perform).

smartfactory

metric	value
rows (N)	23,645
dimension (d)	19
cells	449,255
injected errors	7,093
columns with errors (d_err)	18 of 19
dirty row fraction	0.3000
cell error rate (eps)	0.0158

cell_baseline = 2eps/(1+eps)	0.0311
Rousseeuw prediction	0.2609
denial constraints	18

Detection results

detector	precision	recall	F1	F1 – baseline	#detections	runtime (s)
GreatExpectations (dirty profiled)	1.0000	1.0000	1.0000	0.9689	7,093	0.34
GreatExpectations (clean profiled)	1.0000	1.0000	1.0000	0.9689	7,093	0.33
GreatExpectations (hand written)	1.0000	1.0000	1.0000	0.9689	7,093	0.16
HoloClean	0.0276	0.9862	0.0536	0.0225	253,801	9.70
Raha	0.0171	0.9986	0.0337	0.0026	413,065	169.75

F1 – baseline compares each detector against the trivial detector that flags every cell (cell_baseline = 0.0311). Green beats it by more than 0.02, red is at or below it.

HoloClean — per-constraint ablation

k	constraint (EQ → IQ)	cells	% rows	F1 alone	F1 without	ΔF1	unique TP
7	o_w_blo_voltage, i_w_blo_weg, i_w_bru_weg + 3 more → o_w_bhr_power	22,274	13.5%	0.0358	0.0516	+0.0020	369
6	i_w_blo_weg, o_w_blo_power, i_w_bru_weg + 2 more → o_w_bhl_power	30,648	21.6%	0.0376	0.0516	+0.0020	357
10	i_w_blo_weg, i_w_bru_weg, o_w_hl_voltage + 4 more → i_w_bhr_weg	12,544	6.6%	0.0403	0.0517	+0.0019	360
15	i_w_blo_weg, o_w_blo_power, i_w_bru_weg + 8 more → o_w_hl_power	17,676	6.2%	0.0337	0.0518	+0.0018	417
13	o_w_blo_voltage, i_w_blo_weg, o_w_hl_voltage + 4 more → o_w_hr_voltage	15,440	8.2%	0.0344	0.0519	+0.0017	378
12	o_w_blo_voltage, i_w_blo_weg, i_w_bru_weg + 8 more → o_w_hr_power	16,812	5.9%	0.0316	0.0519	+0.0017	366
9	i_w_blo_weg, o_w_blo_power, o_w_hl_voltage + 3 more → o_w_bhl_voltage	25,067	15.1%	0.0298	0.0521	+0.0016	395
5	o_w_blo_voltage, i_w_blo_weg, i_w_bru_weg + 1 more → i_w_bhl_weg	50,290	42.5%	0.0338	0.0522	+0.0014	282
4	o_w_blo_voltage, i_w_blo_weg, i_w_bru_weg → o_w_hl_voltage	55,432	58.6%	0.0348	0.0525	+0.0011	193
8	o_w_blo_voltage, i_w_blo_weg, i_w_bru_weg + 4 more → o_w_bru_voltage	13,104	6.9%	0.0419	0.0526	+0.0011	217
11	o_w_blo_voltage, i_w_blo_weg, i_w_bru_weg + 5 more → o_w_bhr_voltage	12,672	6.0%	0.0381	0.0526	+0.0010	228
17	o_w_blo_voltage, i_w_bhl_weg, o_w_bhr_power + 5 more → i_w_hr_weg	22,113	10.4%	0.0264	0.0527	+0.0009	383
2	o_w_blo_voltage, i_w_blo_weg → o_w_blo_power	60,453	85.2%	0.0296	0.0531	+0.0005	382
1	o_w_blo_voltage → i_w_blo_weg	47,250	99.9%	0.0283	0.0532	+0.0004	112
14	o_w_blo_voltage, i_w_blo_weg, o_w_hl_voltage + 6 more → i_w_hl_weg	15,560	6.6%	0.0344	0.0532	+0.0004	208
3	i_w_blo_weg → i_w_bru_weg	45,480	96.2%	0.0291	0.0540	-0.0004	7
16	i_w_blo_weg, o_w_blo_power, i_w_bru_weg + 5 more → labels	6,525	3.1%	0.0009	0.0540	-0.0004	4
18	i_w_bru_weg, o_w_bru_voltage, o_w_bhr_voltage + 1 more → o_w_bru_power	73,450	62.1%	0.0284	0.0610	-0.0074	422

All 18 constraints together: P=0.0276, R=0.9862, **F1=0.0536**, flagging **56.5%** of the table. The trivial flag-everything detector scores 0.0311 (+0.0225).

HoloClean touches 19 of the 19 columns. Flagging *every* cell of just those columns — ignoring the constraints entirely and keeping only the column selection — scores F1=0.0311, against HoloClean's 0.0536 (+0.0225 for the row-level filtering the constraints actually perform).

soilmoisture

metric	value
rows (N)	679
dimension (d)	129
cells	87,591
injected errors	26,218
columns with errors (d_err)	128 of 129
dirty row fraction	1.0000
cell error rate (eps)	0.2993
cell_baseline = 2eps/(1+eps)	0.4607

Rousseeuw prediction	1.0000
denial constraints	0

Detection results

detector	precision	recall	F1	F1 – baseline	#detections	runtime (s)
GreatExpectations (dirty profiled)	1.0000	0.0081	0.0161	-0.4446	213	0.26
GreatExpectations (clean profiled)	1.0000	0.0081	0.0161	-0.4446	213	0.26
GreatExpectations (hand written)	1.0000	0.2158	0.3551	-0.1057	5,659	0.12
HoloClean	0.0000	0.0000	0.0000	-0.4607	0	0.40
Raha	0.3672	0.3369	0.3514	-0.1093	24,055	26.03

F1 – baseline compares each detector against the trivial detector that flags every cell (cell_baseline = 0.4607). Green beats it by more than 0.02, red is at or below it.

HoloClean — per-constraint ablation

> No denial constraints for this dataset. HoloClean ran with an empty constraint set, so its 0.0000 is the absence of a constraint set rather than a measurement of the tool. Every FD the FDX profiler found was dropped because its left-hand side contained a (near-)unique column, whose `EQ()` can never match two distinct rows.

adult

metric	value
rows (N)	45,222
dimension (d)	15
cells	678,330
injected errors	1,701
columns with errors (d_err)	1 of 15
dirty row fraction	0.0376
cell error rate (eps)	0.0025
cell_baseline = 2eps/(1+eps)	0.0050
Rousseeuw prediction	0.0370
denial constraints	14

> Every injected error in this copy sits in `income`. A detector that flags any other column is spending its precision on cells that cannot be wrong.

Detection results

detector	precision	recall	F1	F1 – baseline	#detections	runtime (s)
GreatExpectations (dirty profiled)	0.0000	0.0000	0.0000	-0.0050	0	0.33
GreatExpectations (clean profiled)	0.0000	0.0000	0.0000	-0.0050	0	0.33
GreatExpectations (hand written)	0.0000	0.0000	0.0000	-0.0050	0	0.23
HoloClean	0.0025	0.9859	0.0050	0.0000	665,207	26.23
Raha	1.0000	0.0012	0.0023	-0.0027	2	273.45

F1 – baseline compares each detector against the trivial detector that flags every cell (cell_baseline = 0.0050). Green beats it by more than 0.02, red is at or below it.

HoloClean — per-constraint ablation

k	constraint (EQ → IQ)	cells	% rows	F1 alone	F1 without	ΔF1	unique TP
9	education, capital_gain, educational_num + 3 more → income	281,953	89.1%	0.0117	0.0047	+0.0003	111
4	education, capital_gain, educational_num + 1 more → workclass	208,585	92.2%	0.0000	0.0050	-0.0000	0
2	capital_gain → educational_num	90,422	100.0%	0.0000	0.0050	-0.0000	0
1	education → capital_gain	90,444	100.0%	0.0000	0.0050	-0.0000	0
7	education, capital_gain, educational_num + 4 more → gender	238,640	66.0%	0.0000	0.0050	-0.0000	0

5	education, capital_gain, educational_num + 1 more → marital_status	211,715	93.6%	0.0000	0.0050	-0.0000	0
8	education, capital_gain, educational_num + 5 more → native_country	180,387	44.3%	0.0000	0.0050	-0.0000	0
6	education, capital_gain, educational_num + 3 more → race	219,863	69.5%	0.0000	0.0050	-0.0000	0
3	education, capital_gain, educational_num → age	179,656	99.3%	0.0000	0.0050	-0.0000	0
10	education, educational_num, race + 2 more → capital_loss	253,932	93.6%	0.0123	0.0053	-0.0003	17
11	education, educational_num, age + 3 more → hours_per_week	274,393	86.7%	0.0000	0.0054	-0.0003	0
14	capital_gain, marital_status, gender → relationship	175,624	97.1%	0.0000	0.0054	-0.0004	0
12	education, educational_num, workclass + 2 more → fnlwgt	267,498	98.6%	0.0000	0.0054	-0.0004	0
13	workclass, gender, native_country → occupation	180,376	99.7%	0.0000	0.0054	-0.0004	0

All 14 constraints together: P=0.0025, R=0.9859, **F1=0.0050**, flagging **98.1%** of the table. The trivial flag-everything detector scores 0.0050 (+0.0000).

HoloClean touches 15 of the 15 columns. Flagging *every* cell of just those columns — ignoring the constraints entirely and keeping only the column selection — scores F1=0.0050, against HoloClean's 0.0050 (+0.0000 for the row-level filtering the constraints actually perform).